

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

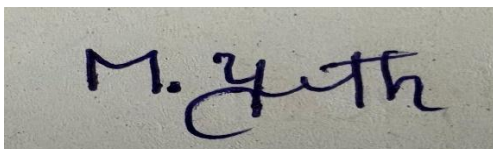
<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M.yeshwanth kumar(192325047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

25-7-26

CO2 AT-1 Assessment

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1. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order),
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

- What is the mean of the data? What is the median?
- What is the mode of the data? Comment on the data modality.
- What is the midrange of the data?
- Can you find (roughly) the first quartile (Q_1) and 3rd quartile (Q_3) of the data?
- Given the five number summary of the data?
- Show a boxplot of the data.

$$a) \text{ mean} = \frac{\sum x_i}{N} = \frac{809}{27} = 29.96 \text{ (Approx)} \approx 30$$

$$= \frac{809}{27} = 29.97 \approx 30$$

$$\text{median: middle value } n=27 \text{ odd} = \frac{27+1}{2} = 14$$

$$14^{\text{th}} \text{ value} = 25$$

mode: 25 & 35 are coming continuously
Bi-modal

midrange:

$$= \frac{\text{max} + \text{min}}{2} = \frac{70 + 13}{2} = \frac{83}{2} = 41.5$$

$$Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25$$

$$\text{median of } Q_1 = 7^{\text{th}} \text{ value} = 20$$

$$Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$$

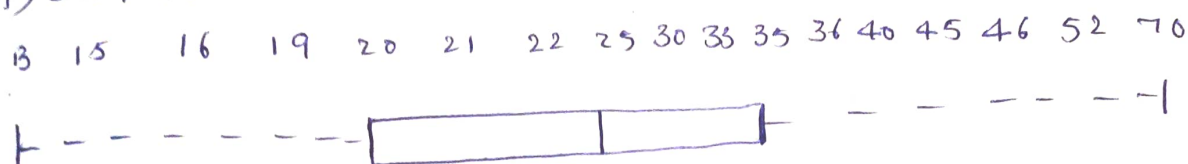
$$\text{median of } Q_3 = \frac{13+1}{2} = 7 \text{th value} = 35$$

e) Five-number Summary

min, Q_1 , median, Q_3 , Max

= 13, 20, 25, 35, 70

f) Boxplot



2. given the following measurements for the variable age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28 standardize the variable by following compute the mean absolute deviation of age

Data for age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

$$n = 10$$

$$\text{Sum} = 18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28 = 330$$

$$= 330 / 10 = 33$$

x age

$|x - \bar{x}|$

18

$$|18 - 33| = 15$$

22

$$|22 - 33| = 11$$

25

$$|25 - 33| = 8$$

42

$$|42 - 33| = 9$$

28

$$|28 - 33| = 5$$

43

$$|43 - 33| = 10$$

33

$$|33 - 33| = 0$$

35

$$|35 - 33| = 2$$

56

$$|56 - 33| = 23$$

28

$$|28 - 33| = 5$$

$$= 15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5$$

$$= 88 \quad \frac{\sum |x - \bar{x}|}{n}$$

$$= \frac{88}{10} = 8.8$$

Mean absolute deviation of age = 8.8

3) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data using a bin depth of 3. Illustrate your steps.

$$\text{mean} = \frac{809}{27} = 29.9 = 30$$

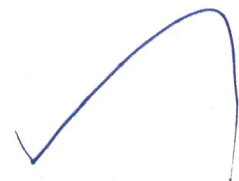
		mean	smoothed values
Bin 1	13, 15, 16	13 14.66	14.67, 14.67, 14.67
Bin 2	16, 19, 20	18 18.33	18.33, 18.33, 18.33
Bin 3	20, 21, 22	21	21, 21, 21
Bin 4	22, 25, 25	24	24, 24, 24
Bin 5	25, 25, 30	26.67	26.67, 26.67, 26.67
Bin 6	33, 33, 35	33.67	33.67
Bin 7	35, 35, 36	35.33	35, 35, 35
Bin 8	40, 45, 46	43.67	40.33, 40.33, 40.33
Bin 9	52, 70	61	56, 56, 56

Smooth mean: 14.67, 14.67, 14.67, 18.33, 18.33, 18.33, 21.00, 21, 21, 24, 24, 24, 26.67, 26.67, 26.67, 33.67, 33.67, 33.67, 35.33, 35.33, 35.33, 43.67, 43.67, 43.67, 43.67, 61, 61.

4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalized by setting min = 0 and max = 1 (b) z-score normalization.

$$n = 5$$

(a) min-max normalization



$$\text{formula} = z' = \frac{x - \min}{\max - \min}$$

$$\max = 1000, \min = 200$$

$$\text{rang} = 1000 - 200 = 800$$

$$200 \quad \frac{200 - 200}{800} = 0$$

$$300 \quad \frac{300 - 200}{800} = 0.125$$

$$400 \quad \frac{400 - 200}{800} = 0.25$$

$$600 \quad \frac{600 - 200}{800} = 0.50$$

$$1000 \quad \frac{1000 - 200}{800} = 1$$

b) step 1: cal mean

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

x	$x - \mu$	$(x - \mu)^2$
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$$\text{Sum of squares} = 90000 + 40000 + 10000 + 10000 + 250000$$

$$= \frac{400000}{5} = 80000$$

$$\text{variance} = \sqrt{S.D} = \sqrt{80000} = 282.84$$

$$z = \frac{x - \mu}{\sigma}$$

200	$\frac{200 - 500}{282.84} = -1.06$
300	$\frac{300 - 500}{282.84} = -0.71$
400	$\frac{400 - 500}{282.84} = -0.35$
600	$\frac{600 - 500}{282.84} = 0.35$
1000	$\frac{1000 - 500}{282.84} = 1.77$

5) Suppose a group of 12 sales price records has been sorted as follows; 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following method

a) equal frequency partitioning b) equal width partitioning

c) clustering

$$n=12$$

a) equal frequency partitioning

Bin 1	5, 10, 11, 13	5 - 13
Bin 2	15, 35, 50, 55	15 - 55
Bin 3	72, 92, 204, 215	72 - 215

b) equal width partitioning

$$\text{range} = \text{max} - \text{min} = 215 - 5 = 210$$

$$\text{width} = 210 / 3 = 70$$

Bin 1	5 - 74	values 5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75 - 144	92
Bin 3	145 - 215	204, 215

c) clustering

Bin 1	5, 10, 11, 13, 15	Cluster 1: close value 5 - 15
Bin 2	35, 50, 55, 72, 92	Cluster 2: 35 - 92
Bin 3	204, 215	Cluster 3: outliers 200+

6) Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.9	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	32.5	28.8	33.4	30.2	34.1	32.9	41.2	35.9

age: 23, 33

sum = 856

mean = $856/18 = 47.56$

median = avg of 9th + 10th = $\frac{50+51}{2} = 50.5$

$$\text{variance} = \frac{\sum (x - \mu)^2}{n} = \frac{2236.89}{18}$$

b) Boxplot - 5 number Summary

min = 23, $Q_1 = 33$, median = 51, $Q_3 = 57$ max = 61

$Q =$ median of lower 9th = 39

1

2

c) Scatter plot and Q-Q Plot

mean
25/11/16

Scatter Plot Observation

The point an upward trend

As increase, Body fat (%) also tends to increase

Q-Q Plot observation

for both variables

Points lie approximately normally distributed

d) Z-score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

x = original value

μ = mean

σ = sample standard deviation

Age	Z-score	% fat	Z-score
23	-1.774	9.5	-2.084
23	-1.774	26.5	-0.247
27	-1.471	7.8	-2.267
27	-1.471	17.8	-1.187
39	-0.563	31.4	0.283
41	-0.412	25.9	-0.312
47	0.042	27.4	-0.149
49	0.193	27.2	-0.171
50	0.269	31.2	0.261
52	0.420	34.1	0.629

54	0.572	28.8	0.629
54	0.723	34.4	1.482
56	0.799	30.2	0.002
57	0.874	34.1	0.499
58	0.874		0.153
58	0.874	32.9	0.578
60	1.025	41.9	0.445
61	1.101	35.5	1.342
			0.747

e) Pearson's Correlation coefficient

$$r = 0.8176$$

Interpretation

$$\text{Correlation Coefficient } (r) = 0.8176$$

Since $r > 0$ the variables are positively correlated.

0.8176 is close to 1. relationship is strong positive.

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